AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2024

Mathematics Paper II

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are NINE questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. A formulae list is provided on page 2 and 3. You may refer to it during the paper, if you wish.

6. You may use a scientific calculator if you wish.

List of Formulae

Note:

- All symbols used in the formulae have their usual meaning.

Functions and Limits

$$\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1 \qquad \lim_{x\to 0}\frac{\sqrt{x+a}-\sqrt{a}}{x}=\frac{1}{2\sqrt{a}} \qquad \lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}$$

$$\lim_{x\to 0}(1+x)^{\frac{1}{x}}=e \qquad \lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x=e \qquad \lim_{x\to 0}\frac{a^x-1}{x}=\log_e a$$

Differentiation

$$\frac{\Delta y}{\Delta x}=\frac{f(x+h)-f(x)}{h} \qquad \frac{d}{dx}(\sec x)=\sec x\tan x \qquad \frac{d}{dx}x^n=nx^{n-1} \qquad \frac{d}{dx}(\sin x)=\cos x$$

$$\frac{d}{dx}(\cos x)=-\sin x \qquad \frac{d}{dx}(\tan x)=\sec^2 x \qquad \frac{d}{dx}(\cot x)=-\operatorname{cosec}^2 x \qquad \frac{d}{dx}(\operatorname{cosec} x)=-\operatorname{cosec} x\cot x$$

$$\frac{d}{dx}\left[\sin^{-1}x\right]=\frac{1}{\sqrt{1-x^2}},\ x\in(-1,1) \qquad \frac{d}{dx}\left[\cos^{-1}x\right]=-\frac{1}{\sqrt{1-x^2}},\ x\in(-1,1) \qquad \frac{d}{dx}\left[\tan^{-1}x\right]=\frac{1}{x^2+1},\ x\in R \qquad \frac{d}{dx}\left[\sec^{-1}x\right]=\frac{1}{|x|\sqrt{x^2-1}},\ x\in[-1,1]$$

$$\frac{d}{dx}\left[\operatorname{cosec}^{-1}x\right]=-\frac{1}{|x|\sqrt{x^2-1}},\ x\in[-1,1] \qquad \frac{d}{dx}\left[\cot^{-1}x\right]=-\frac{1}{1+x^2},\ x\in R \qquad \frac{d}{dx}(a^x)=a^x\ln a \qquad \frac{d}{dx}(\log_a x)=\frac{1}{x}\times\frac{1}{\ln a}$$

$$\frac{d}{dx}(\sinh x)=\cosh x \qquad \frac{d}{dx}(\cosh x)=\sinh x \qquad \frac{d}{dx}(\tanh x)=\operatorname{sech}^2 x$$

$$\frac{d}{dx}(\operatorname{cosech} x)=-\coth x\operatorname{cosech} x \qquad \frac{d}{dx}(\operatorname{sech} x)=-\tanh x\operatorname{sech} x \qquad \frac{d}{dx}(\coth x)=-\operatorname{cosech}^2 x$$

$$\frac{d}{dx}\left[f(x)\times g(x)\right]=f(x)\frac{d}{dx}\left[g(x)\right]+g(x)\frac{d}{dx}\left[f(x)\right] \qquad \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right)=\frac{g(x)\frac{d}{dx}\left[f(x)\right]-f(x)\frac{d}{dx}\left[g(x)\right]}{\left[g(x)\right]^2}$$

Maclaurin Series $f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+\ldots+\frac{f^n(0)}{n!}x^n+\ldots$

Taylor's Series $f(x)=f(a)+f'(a)(x-a)+\frac{f''(a)}{2!}(x-a)^2+\frac{f'''(a)}{3!}(x-a)^3+\frac{f^{iv}(a)}{4!}(x-a)^4+\ldots+\frac{f^n(a)}{n!}(x-a)^n+\ldots$

Integration

$$\int f'(x)dx=f(x)+c \qquad \int[f(x)]^n f'(x)dx=\frac{[f(x)^{n+1}]}{n+1}+c,\quad (n\neq -1) \qquad \int f(x)g'(x)dx=f(x)g(x)-\int f'(x)g(x)dx$$

$$\int\frac{f'(x)}{f(x)}dx=\ln f(x)+c \qquad \int\operatorname{cosec} x\,dx=\ln|\operatorname{cosec} x-\cot x|+c \qquad \int\sec x dx=\ln|\sec x+\tan x|+c$$

Plane Analytical Geometry (Straight Line)

$$d=\frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}} \qquad y-y_1=\frac{y_2-y_1}{x_2-x_1}(x-x_1) \qquad \left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$$

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Point of internal division

$$\left(\frac{k_1x_2+k_2x_1}{k_1+k_2},\ \frac{k_1y_2+k_2y_1}{k_1+k_2}\right)$$

Point of external division

$$\left(\frac{k_1x_2-k_2x_1}{k_1-k_2},\ \frac{k_1y_2-k_2y_1}{k_1-k_2}\right)$$

$$\frac{x-x_1}{\cos\alpha}=\frac{y-y_1}{\sin\alpha}=r$$

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$\frac{x}{a}+\frac{y}{b}=1$$

$$x\cos\alpha+y\sin\alpha=p$$

$$y=mx+c$$

$$y-y_1=m(x-x_1)$$

$$\theta=\tan^{-1}\left[\pm\frac{m_2-m_1}{1+m_1m_2}\right]$$

$$\theta=\tan^{-1}\left[\frac{2\sqrt{h^2-ab}}{a+b}\right]$$

$$\begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}=0$$

$$\Delta=\frac{1}{2}\begin{vmatrix}x_1 & y_1 & 1\\ x_2 & y_2 & 1\\ x_3 & y_3 & 1\end{vmatrix}$$

Circles

$$(x-h)^2+(y-k)^2=r^2$$

$$x^2+y^2+2gx+2fy+c=0$$

Equation of normal to a circle

$$(y-y_1)(x_1+g)=(x-x_1)(y_1+f)$$

Equation of tangent to a circle

$$xx_1+yy_1+g(x+x_1)+f(y+y_1)+c=0$$

Length of tangent to a circle from a point (x_1, y_1), $l=\sqrt{x_1^2+y_1^2+2gx_1+2fy_1+c}$

Parabola

$$x^2=4ay \qquad (x-h)^2=4a(y-k) \qquad y^2=4ax \qquad (y-k)^2=4a(x-h)$$

Ellipse

$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1,\ a>b \qquad b^2=a^2(1-e^2) \qquad c=ae \qquad \frac{(y-k)^2}{a^2}+\frac{(x-h)^2}{b^2}=1,\ a>b$$

Hyperbola

$$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1 \qquad b^2=a^2(e^2-1) \qquad c=ae \qquad \frac{(y-k)^2}{a^2}-\frac{(x-h)^2}{b^2}=1$$

Translation and Rotation

$$x=X+h,\ y=Y+k \qquad X=x\cos\theta+y\sin\theta,\ Y=y\cos\theta-x\sin\theta \qquad \tan 2\theta=\frac{2h}{a-b}$$

Vectors

$$\underline{u}.\underline{v}=|\underline{u}||\underline{v}|\cos\theta \qquad \text{Area of a triangle}=\frac{1}{2}|\underline{u}\times\underline{v}| \qquad \underline{u}\times\underline{v}=\left(|\underline{u}||\underline{v}|\sin\theta\right)\hat{\underline{n}} \qquad \underline{r}=\frac{q\underline{a}+p\underline{b}}{p+q}$$

$$\left|\overrightarrow{P_1P_2}\right|=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2} \qquad \frac{1}{6}(\underline{u}\times\underline{v}).\underline{w}=\frac{1}{6}[\underline{u}\ \underline{v}\ \underline{w}] \qquad \underline{u}.(\underline{v}\times\underline{w})=\begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}$$

Some Trigonometric Identities

$$\cos(\alpha\pm\beta)=\cos\alpha\cos\beta\mp\sin\alpha\sin\beta$$

$$\sin(\alpha\pm\beta)=\sin\alpha\cos\beta\pm\cos\alpha\sin\beta$$

$$\cos P+\cos Q=2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2}$$

$$\cos P-\cos Q=-2\sin\frac{P+Q}{2}\sin\frac{P-Q}{2}$$

$$\sin P+\sin Q=2\sin\frac{P+Q}{2}\cos\frac{P-Q}{2}$$

$$\sin P-\sin Q=2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$$

$$\sin 2\alpha=2\sin\alpha\cos\alpha$$

$$\cos 2\alpha=\cos^2\alpha-\sin^2\alpha$$

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Q.1. (Total 3 Marks)

i. If $f(x) = 3x^2 - 5x + 1$ and $g(x) = x + 1$, then find $g \bullet f$. (1 Mark)

ii. Evaluate $\lim_{x \to \infty} \frac{x^2 - 2x + 5}{x^2 - 4}$ when $x \neq \pm 4$. (2 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

Q.2. (Total 10 Marks)

a. (5 Marks)

Using logarithms, differentiate the given function with respect to x.

$$y = \frac{(x+1)(x-2)}{x+3}$$

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

b. (5 Marks)

Find $\frac{dy}{dx}$ for the given function.

$x = \sin t$, $y = \cos \pi t$

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

c. (5 Marks)

Find the Taylor series expansion of the given function about the given value up to four terms.

$f(x) = \sin x,\ a = \frac{\pi}{4}$

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

Q.3. (Total 10 Marks)

a. Using integration by substitution, evaluate $\int \frac{dx}{\sqrt{x^2-9}} = \ln\left(\frac{x}{3} + \frac{\sqrt{x^2-9}}{3}\right) + c$. (5 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

b. Apply the method of partial fraction to evaluate the given integral. (5 Marks)

$$\int \frac{x}{(x-4)(x+5)}\,dx$$

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

c. (5 Marks)

An object is dropped from the top of a tower, vertically downwards, under the influence of gravitational force.

Taking gravitational acceleration as 10 m/s^2, find the

i. expression for velocity *v* after *t* seconds of falling. (3 Marks)

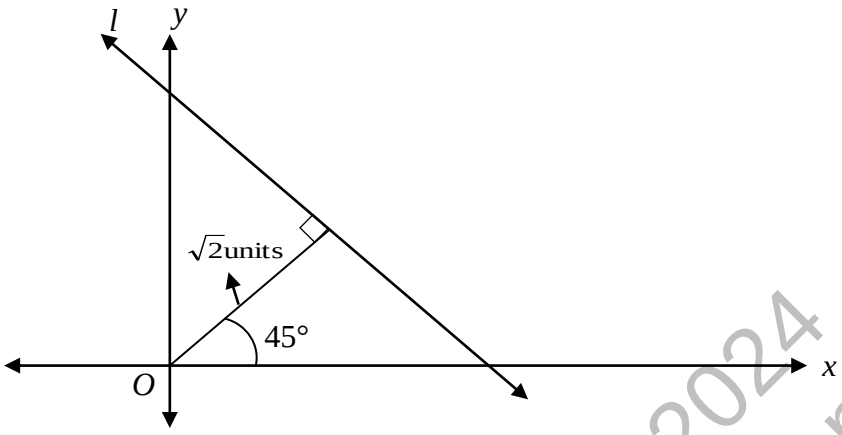
ii. velocity after 2 seconds of the fall. (1 Mark)

iii. hence, find the expression for displacement. (1 Mark)

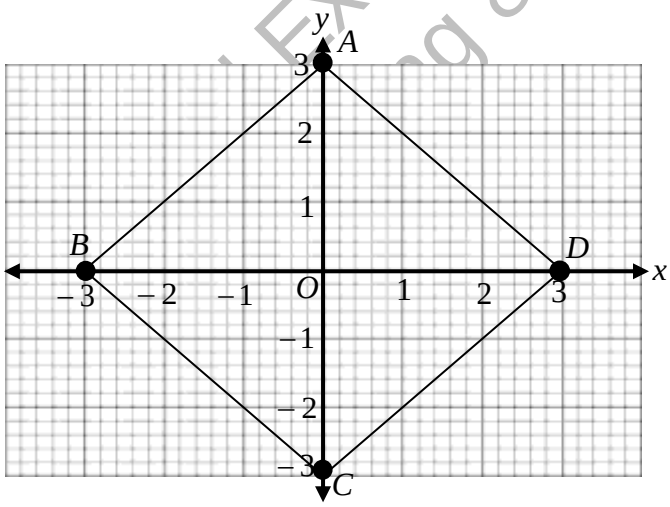
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Q.4. (Total 4 Marks)

i. Find the equation of the line l in the given situation by using the normal form. (2 Marks)



ii. Find the area of the quadrilateral $ABCD$ with the given vertices. (2 Marks)



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Q.5. (Total 4 Marks)

The given table shows the production time, inspection time and profit per unit of the two manufactured cars *A* and *B*.

Types	Production Time in Hours (Per Unit)	Inspection Time in Hours (Per Unit)	Profit in Rs 10,000 (Per Unit)
A	4	2	2
B	5	3	3
Maximum	18	8	

Based on this linear programing problem, construct

i. constraints for the values of unknowns (x and y). (1 Mark)

ii. constraints for the number of hours of production and inspection. (2 Marks)

iii. an objective function to maximise the profit. (1 Mark)

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Q.6. (Total 4 Marks)

Find the centre of the circle passing through the points (0, 0) and (3, –2) and having centre at the line $3x + y = 10$.

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

Q.7. (Total 8 Marks)

a. Find the equation of the normal to the parabola $y^2 = -8x$ at (2, 2) in the standard form. (4 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

b. An ellipse with centre $(0,0)$ has vertices $(\pm 5,0)$ and foci $(\pm 4,0)$. Find the

i. equation of the ellipse. (2 Marks)

ii. eccentricity. (1 Mark)

iii. equation of directrices. (1 Mark)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

c. The vertices of a hyperbola are $(15, 1)$ and $(-1, 1)$. If the length of its conjugate axis is 12 units, then find the equation of the hyperbola in standard form. (4 Marks)

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Q.8. (Total 4 Marks)

Find the equation of the hyperbola $x^2 - y^2 = 1$ with respect to new axes obtained by rotation of axes about the origin through an angle of $\frac{\pi}{4}$ radians.

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(ATTEMPT EITHER PART a OR PART b OF Q.9.)

Q.9. (Total 3 Marks)

a. Find the value of *x* for which the two vectors $\vec{A} = x\hat{i} - 5\hat{j}$ and $\vec{B} = \hat{i} - 3\hat{j} + 5\hat{k}$ have same magnitudes.

b. Find the angle between the vectors $m = 3\hat{i} - \hat{j} - 3\hat{k}$ and $n = \hat{i} - \hat{j}$.

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END OF PAPER

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